

Max Ratcliff

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Summary

Applied Mathematics & Computer Engineering undergraduate at UC Santa Cruz with experience building data and machine-learning systems for scientific research. At **NASA JPL**, developing a pipeline that organizes Mars imagery and scientific context for a vision-language model. Seeking machine learning engineering opportunities and software engineering roles involving data-intensive or scientific systems.

Education

University of California, Santa Cruz, Santa Cruz, CA Expected June 2027
B.S Double Major: Applied Mathematics & Computer Engineering (Systems Concentration) 3.6/4.0

Relevant Coursework:

- **Math & ML:** Scientific ML, AI Research, ML & Data Mining, Linear Algebra, ODEs/PDEs, Vector Calculus.
- **Systems & CS:** Logic Design (Verilog), Computer Networking, Computer System Design, Operating Systems, Data Structures.

Skills

Programming: Python, C, C++, Verilog, RISC-V ASM, MATLAB, Java, JavaScript, Fortran 90, HTML/CSS, LaTeX

Libraries & Frameworks: PyTorch, JAX, NumPy, SciPy, Pandas, Matplotlib, FastAPI, React, Node.js, Docker

Tools & Platforms: Git, Unix/Linux, Google Cloud, Firebase, SupaBase

Experience

Robotics ML Intern, NASA Jet Propulsion Laboratory June 2026 — Present

- Built an end-to-end Python pipeline to collect and process **5TB+** of raw Mars data, including **200,000+** surface images and associated science or mission annotations, for a domain-specific vision-language model.
- Developed a multimodal meeting-ingestion pipeline with FFmpeg, local Whisper, OCR, and OpenCV ORB matching to associate transcript segments with slides; using the resulting corpus to connect Mars targets, images, and scientific context for LLaVA-style model training.

Software Engineering Intern, AMI

April 2026 — June 2026

- Developed an LLM orchestration platform (LiteLLM-based) to unify prompt management, key privacy, and usage guardrails.
- Wrote API documentation for centralized prompt management endpoints, streamlining model integration across internal teams.

Undergraduate Researcher, UC Santa Cruz

Jan 2026 — April 2026

- Researched aligning Spiking Neural Networks (SNNs) with biological manifolds using Joint-Embedding Predictive Architectures (JEPA).
- Implemented cloud training pipelines using snntorch and LeJEPA, validating neural alignment via Canonical Correlation Analysis (CCA) with high-power GPUs via Google Colab.
- Developing interpretable neural circuit motifs by filtering noise through JEPA's predictive objectives, bridging deep learning and biological plausibility.

SciAI Pathways Scholar, UC Santa Cruz (ARO Lab)

June 2024 — Present

- Developing a multimodal wound-healing model combining wound images (ResNet-50) with RNA genomic data (MLP) in PyTorch.
- Mentoring first-year researchers on deep learning implementation and exploring embedding architectures for small-sample optimization.
- Engineered a fully automated data pipeline in Python using CellPose to process 5,000+ microscopy images with 99% accuracy and 0.3ms latency; awaiting experimental results for publication.

Resident Assistant, UC Santa Cruz

Sept 2024 — Present

- Direct leadership resource for 60 residents and 24/7 on-call emergency lead for a community of 2,500+ students; managing conflict resolution and crisis intervention.
- Mentor new RAs on effective conflict resolution and community building strategies.

Senior Lifeguard & Instructor (Supervisor), City of Pasadena

Summers 2020 — 2025

- Acted as third-in-command for daily pool operations, supervising staff and managing safety for 300+ daily swimmers.
- Instructed 2,000+ students in Red Cross curriculum, teaching water safety and swimming skills at all levels.

Projects

Neural Network Chess Engine

chess.ratcliff.cc

- Built a hybrid engine pairing Minimax search with a dual-head ResNet, achieving a 1647 ELO rating; trained on 160 million positions using A100 GPUs.
- Optimized inference via policy pruning, achieving 15,000 nodes/sec on an M2 chip for 6-ply depth play.

Autoregressive Weather Model

weather.ratcliff.cc

- Trained a convolutional VAE on 5 years of historical data for sparse sensor reconstruction (0.00032 MSE loss).